Sebastian Mutuku

Senior Backend Engineer · Core Banking · Android · HAPI FHIR

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PROFESSIONAL SUMMARY

Senior Backend Engineer with **8+ years** of experience building high-volume banking, payment and enterprise backend systems. Strong expertise in Java, Spring Boot, Temenos T24, REST APIs and Kubernetes. Proven record of delivering production-grade financial systems, modernising legacy platforms and deploying scalable services in regulated environments. Open-source contributor: published a Java artifact to Maven Central.

CORE SKILLS

Backend	Java, Kotlin, Python, Spring Boot, Spring Security, Quarkus, Ktor, Django, Scala/Play, REST, gRPC, GraphQL
Core banking	Temenos T24, Finacle, ISO 8583, integrations, card processing
Mobile	Android (Kotlin, Jetpack Compose, Hilt, Retrofit), React Native, Flutter, HAPI/Google FHIR SDK
Data & messaging	PostgreSQL, Oracle, MySQL, SQL Server, MongoDB, CockroachDB, TimescaleDB, Redis, Kafka, RabbitMQ
DevOps & cloud	Docker, Kubernetes, OpenShift, Istio, Terraform, Ansible, GitHub Actions, HashiCorp Vault, WSO2, Prometheus, Grafana Loki, SonarQube, AWS, GCP, Azure
Networking	C++, socket programming, TCP/IP, UDP, DNS, DHCP, multicasting, routing protocols, network security

EXPERIENCE

Senior Applications Developer — KCB Bank Group

Oct 2024 – Present

Lead developer for backend systems, API development and core banking integration.

- Spearheaded a microservices architecture for banking applications.
- Reduced transaction processing time by **30%** through backend optimisation.
- Migrated core banking applications to Kubernetes and OpenShift.
- Introduced a configuration server and HashiCorp Vault for secure configuration management.
- Built an ATM card processing system, an Agency Banking solution and a Bancassurance platform, all integrated with T24.
- Mentored junior developers and led code reviews.

Senior Software Engineer — ONA Kenya Ltd

Nov 2021 – Sep 2024

Contributed to the HAPI FHIR SDK and built geospatial Android applications.

- Led development of a mobile health app used by over **120,000** healthcare workers.
- Enhanced the HAPI FHIR SDK with new FHIR resources and better performance.
- Improved application data-processing speed by **25%**.
- Delivered geospatial solutions for health and infrastructure clients.
- Integrated FHIR-based applications with existing health information systems; trained clients on FHIR standards.

Applications Developer — KCB Bank Group

Sep 2018 – Oct 2021

Developed and maintained banking applications, focusing on backend services and integrations.

- Built backend services with Java and Spring Boot, integrated with T24 and Finacle.
- Delivered the Checkbook ordering system for KCB Tanzania and a card ordering system for KCB Kenya.
- Enhanced a system that posts external transactions to T24 via REST APIs.
- Improved reliability and performance through refactoring; worked across the full SDLC.

Software Engineer Intern — SimbaTech Solutions Ltd

Apr 2018 – Aug 2018

Developed Finacle modules and provided technical support.

- Built a credit card module on Finacle core banking for Victoria Commercial Bank.
- Provided technical support and took part in code reviews and knowledge-sharing sessions.

SELECTED PROJECTS

- **Agency Banking Solution** — Android, Java, Spring, Oracle, T24, OpenShift — new features, security and performance for high transaction volumes.
- **Bancassurance Platform** — Java, Spring Boot, Kotlin, PostgreSQL, T24 — unified banking and insurance services.
- **ATM Card Processing System** — Java, Spring, Kotlin, PostgreSQL, ISO 8583, T24 — card issuance, activation and transactions.
- **WHO ANC & Child Immunization apps** — Kotlin, Jetpack Compose, Google FHIR SDK, Firebase — implementing WHO guidelines for health workers.
- **Checkbook Ordering & School Gear Payments** — Spring Boot / Quarkus, Kafka, RabbitMQ, WSO2, Istio — backends for KCB Tanzania and KCB Rwanda / BPR.

EDUCATION

B.Sc. Electronics — Computer Engineering

LINKS

Portfolio: SebaMutuku.github.io/sebamutuku-portfolio | GitHub: github.com/SebaMutuku